Common knee injuries and how you can avoid them

by Dan Volpini, MD
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From young athletes to older adults, knee injuries are one of the most common orthopedic injuries we treat. When you're on your feet, your knees take on the load of your weight. This pressure and stress can make your knees more prone to injury, especially compared to other joints like the wrist or elbow.

While not every knee injury can be avoided, you can reduce your risk or decrease the severity of an injury when it happens. Here we'll cover three common types of knee injuries, who is at risk and the basic steps you can take to protect your knees.

ACL injuries

ACL injuries—which are injuries to the knee's anterior cruciate ligament—make up about half of all knee injuries. The ACL is the main ligament in the center of your knee, and it provides interior stability.

We see ACL injuries most often in athletes who do a lot of pivoting or change of direction, such as in soccer, volleyball, football or basketball. If the athlete makes a quick direction change and the muscles around the knee can't take on the pressure and protect the ACL, it can tear.

Female athletes are particularly at risk for an ACL injury. Their anatomy and landing mechanics in sports like basketball and volleyball can predispose them to it. The ratio of female ACL injuries to male ACL injuries is almost 5 to 1.

Meniscus injuries

Another common knee injury is a meniscus tear. The meniscus is cartilage that creates a cushion between the two bones in your knee—the femur and tibia. Motions that twist the knee from side to side can cause this cartilage to tear.

Two groups are more likely to tear their meniscus. First, we often see meniscus tears in younger athletes as the result of a twisting motion during sports. It's also common for young athletes with an ACL tear to have a meniscus tear at the same time. This happens in about half of all ACL cases.

The other group is older adults with degenerative meniscus tears. As you get older, you lose water content from the cartilage, causing it to become more fragile. This wear and tear can lead to a meniscus injury when an older adult puts stress on the knee.

Overuse injuries

The third major type of knee injury is an overuse injury, which involves inflammation in the knee. For example, runner's knee is an overuse injury that causes inflammation and pain around the kneecap. Tendinitis occurs due to overuse of a tendon, which is a cord that connects muscles and bones.

People who do repetitive activities, like runners and athletes, are more at risk for this type of injury. Overuse injuries may also be more likely in those who spend too much time on their feet or have poor posture.

Typically, overuse injuries don't require major intervention. We can treat them by allowing the knee to rest. At-home treatment includes RICE: rest, ice, compression and elevation.

3 ways to help prevent knee injuries

If you're in one of the groups more prone to a knee injury, you can take steps to reduce your chances of an injury. Three of the most important things you can do to help prevent knee injuries include:

- **Do resistance exercises** that build up the musculature around the knee, including the quads and hamstrings. These muscles can help take the load and pressure off of your knee.
- **Always stretch** before activity. Stretching can help improve the flexibility of the muscles around your knee.
- **Pay attention to form** and posture when you're exercising. If you run or play sports with poor form, it can cause more harm than good.

Protecting your knee health

Overall, the best thing you can do for your knee health is to live a healthy lifestyle. Focus on weight control, and exercise and listen to your body. If your knee is hurting, your body is telling you something isn't right, and you don't want to overlook chronic knee pain.

Some knee injuries will improve with ice, rest and anti-inflammatory medication, but others require surgery. If you have an injury that isn't getting better in 2-3 days or has swelling and instability, it merits attention from a doctor. With the right treatment, most people with knee injuries go on to regain function and have a positive outcome.

About the Author

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